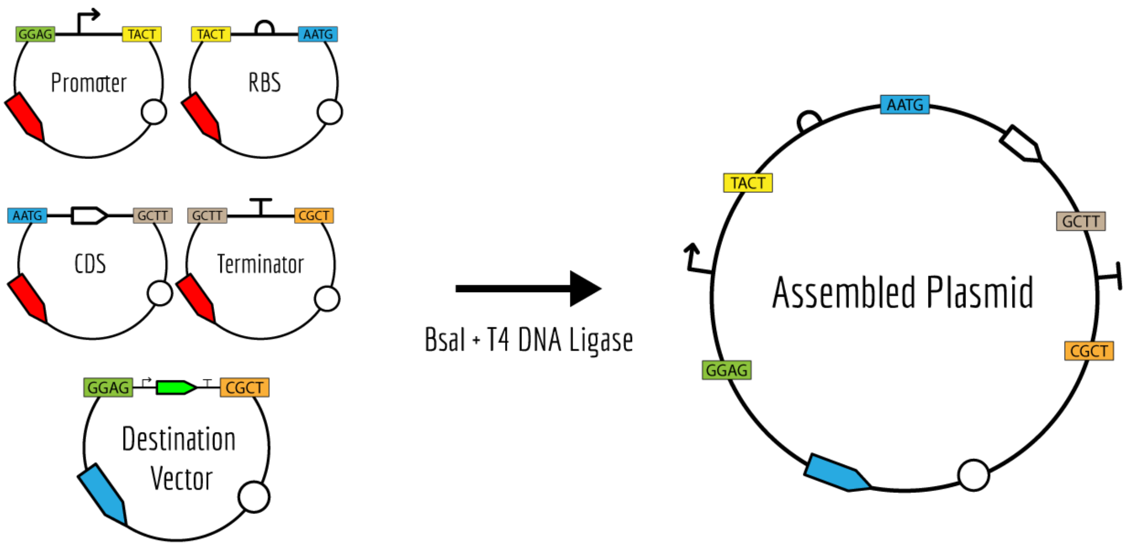
procedural writeup

Notes:

- Use TWO cassettes
- General golden gate assembly order: promoter, rbs, cds, terminator
- JUMP vector from kit will serve as backbone
- BtSULT needs PAPS as the sulfate donor in order to convert LCA (secondary bile acid) to sulfated LCA (product)

Operon layout:
LR (ligation sites)
Promoter
LR
RBS
LR
Gene/CDS
LR
2X Terminator
50+ BP NOISE
LR
Next promoter
LR
Next RBS
LR
Next CDS
LR
Next 2X terminator

Assembly -



Assignment: create a **hierarchical Golden Gate build** in Benchling, using the iGEM/Distribution Kit parts as standardized modules, and then combining those modules into a JUMP backbone to make the final operon construct. Golden Gate is ideal for this because Type IIS enzymes cut outside their recognition sites, creating custom overhangs so multiple DNA parts can be assembled in one direction in one reaction.

Cassette 1: BtSULT

- Use one Anderson promoter upstream of BtSULT.
- Use one RBS from the kit.
- Use the BtSULT CDS.
- Use double terminator downstream

Cassette 2: PAPS supply

- Use one Anderson promoter upstream of KIATPSL + PcAPSK
- Use one RBS from the kit.
- Use the KIATPSL + PcAPSK CDS
- Use double terminator downstream

1) Backbone checklist

- Is a JUMP vector or other kit-approved destination backbone.
- Is compatible with the same assembly standard as your inserts.
- Can accept two cassettes in one construct.
- Has the correct selection marker for your workflow.
- Is listed as a valid recipient/backbone rather than an insert part.
- Matches the assembly level you are building first and the final level you want.

2) Promoter checklist

- Collection/type: Anderson promoter.
- Host compatibility: works in *E. coli* / your chassis.
- Strength: weak, medium, or strong, depending on how much expression you want.
- Role: constitutive promoter, unless your project specifically needs inducible control.
- Assembly compatibility: correct Golden Gate format and flanking sites.
- Independence: no unwanted coding sequence or terminator attached unless it is a composite part you intentionally want.

3) RBS checklist

- Part role: RBS or translation initiation region.
- Assembly format: compatible with the same Golden Gate system.
- Strength: strong, medium, or weak depending on desired translation level.
- Context: intended for bacterial expression.
- Boundary cleanliness: no promoter/CDS mixed in unless it is a composite part.
- Consistency: use the same general style of RBS for both cassettes unless you are deliberately tuning expression.

4) CDS checklist

- Exact gene identity: BtSULT for cassette 1; KlATPSL + PcAPSK for cassette 2.
- Correct orientation: forward and ready to be placed after the RBS.
- No extra regulatory parts unless the part is intentionally composite.
- Codon optimization status: if available, prefer the version designed for your host.
- Assembly compatibility: flanking sites/overhangs that place it correctly between RBS and terminator.
- Completeness: the CDS should match the full protein-coding region you need.

5) Terminator checklist

- Part role: terminator.
- Strength/format: a standard terminator or double terminator if your plan requires it.
- Placement: downstream of the CDS.
- Compatibility: matches the same assembly standard as the rest of the cassette.
- Isolation: ensures transcription stops before the next cassette.
- No accidental promoter overlap unless it is a designed composite part.

6) Cassette 1 checklist: BtSULT

- One Anderson promoter.
- One bacterial RBS.
- BtSULT CDS.
- One double terminator downstream.
- Compatible Golden Gate boundaries.
- Good expression balance for enzyme production.

7) Cassette 2 checklist: PAPS supply

- One Anderson promoter.
- One bacterial RBS.
- KlATPSL + PcAPSK CDS.
- One double terminator downstream.
- Compatible Golden Gate boundaries.
- Sufficient expression to support PAPS production without excessive burden.

8) Whole-operon checklist

- Exactly two cassettes.
- Cassette order is correct.
- Each cassette follows promoter → RBS → CDS → terminator.
- The JUMP backbone is compatible with the whole construct.
- The parts do not create unwanted fusion junctions.
- The promoter strengths make biological sense for the pathway.

Important information from kit

- Part name.
- Part role.
- Collection.
- Assembly format.
- Flanking restriction site.
- 5' and 3' overhangs.
- Any notes that say “Level 0,” “Level 1,” or “composite.”

parts from kit

Backbones:

- JUMP/plasmid vector (pJUMP29-1A(sfGFP))

Promoter:

-

steps

- Pick the backbone first.
- Choose one promoter per cassette (Anderson promoter)
- Add an RBS after each promoter.
- Place the coding sequence after the RBS.
- Add a terminator at the end of each cassette.
- Verify that every fragment has the right Golden Gate overhangs so they assemble in the intended order.
- Check the distribution kit for matching part IDs and well locations.
- Build the final design in Benchling and export the assembly map for the wet lab team.

promoter → RBS → gene → terminator

Your writeup should explain:

- how Golden Gate works,
- which kit parts you selected,
- why those promoters were chosen,
- what backbone you used,
- and how the pieces were ordered into a final construct.

terms

Golden Gate Assembly (GGA) is an extremely powerful modular assembly technique in synthetic biology that allows for the efficient and precise assembly of multiple DNA fragments into a single construct. It can be used to create complex DNA constructs, such as expression plasmids, circuits or gene clusters.

The power of Golden Gate Assembly permits the creation of off the shelf libraries of DNA parts that can be used for single construct assembly or multiplexed assembly of many constructs. This technique also shines in its ease of multiplexing using automation.

Golden Gate Assembly comprises a number of constituent elements: Type IIS Restriction Enzymes, Parts and Plasmids, and Assembly Standards.

Type IIS restriction enzymes bind to their respective recognition sites, but then cleave the DNA at defined positions away from the recognition sites. After cleaving, single-stranded overhangs remain.

Type IIS recognition sites are also not palindromic, which means that the two strands of the DNA it comprises do not have the same sequence. This permits the flanking of a DNA part, ex. a promoter or gene, with the same recognition site in opposing directions such that the part can be cleaved from its backbone, removing the recognition sites but leaving behind those single-stranded overhangs.

- The most commonly used Type IIS restriction enzymes in Golden Gate Assembly are the four base pair cutters BsaI, BbsI, BsmBI, and the three base pair cutter SapI, or isoschizomers of these enzymes.
- Among these, BsaI is most often used for assembling basic parts into a transcription unit.

Anderson promoters are a family of constitutive promoters that differ in strength, so they let you tune how much each gene is expressed. In your project, promoters may be used to control:

- The enzyme gene expression level.
- The transporter expression level.

- Different operon modules if your team splits the pathway into multiple transcription units.

JUMP vectors are modular cloning backbones often used for Golden Gate-style assembly, especially when teams want a standardized backbone that works with interchangeable inserts.

The distribution kit is where you look for standardized parts to reuse instead of building everything from scratch. Your task is to identify which kit entries match:

- promoters,
- RBSs,
- terminators,
- coding sequences,
- and backbones

In your PCS probiotic design, BtSULT needs PAPS as its sulfate donor to convert LCA into sulfated LCA.

Hierarchical Levels of Assembly

The platform details how genetic circuits are built hierarchically:

- **Level 0 (Basic Parts):** Individual, discrete genetic elements (e.g., a single promoter, an RBS, or a protein-coding gene) stored in a storage vector.
- **Level 1 (Transcriptional Units):** The assembly of Level 0 parts together (Promoter + RBS + CDS + Terminator) into a complete functional unit using **BsaI**.
- **Level 2+ (Multi-Transcriptional Units):** Combining multiple Level 1 transcriptional units together to build complex, multi-gene pathways or circuits, often utilizing **SapI** or **BsmBI**.

Table 1. With definitions of level 0, level 1 and level 2 plasmids.

Level 0: constructs are the fundamental parts of a transcriptional unit each placed into a level 0 plasmid. A transcriptional unit is a series of four level 0 units that can be transcribed into an mRNA. Alone, these parts often can do little transcriptionally or translationally.
Level 1: constructs are transcriptional units placed into level 1 plasmids. A transcriptional unit is composed of a promoter, 5' untranslated region, a coding domain sequence and a terminator.
Level 2: constructs are four transcriptional units placed into a level 2 plasmid vector.
Level 3: constructs are 4 main modules of level 2 plasmids (each with four transcriptional units)placed into a level 1 plasmid to create a 16 transcriptional unit construct.

Documentation

[JUMP guide](#)
https://technology.igem.org/distribution-kit?tab=handbook
https://technology.igem.org/technologies/golden-gate

7/6/2026; 6:45 PM - 9:20 PM; Humza, Kylie, Kyle, Tiger
Meeting Goals:

- Conduct research on L1 to L2 JUMP Plasmid assembly
    - Read JUMP guide + original paper
    - Discuss gene assembly method and protocol
    - Select backbones and potential workflow

Research and Meeting Notes:

JUMP Guide Notes:

Cargo Module:
"
1. An upstream module with two AarI restriction sites.
2. A main module containing BsaI/BsmBI sites. (Level 0 JUMP plasmids contain BsmBI, level 1 contains BsaI sites and level 3 contains BsmBi sites)
3. A downstream module containing BbsI sites.

Joint Universal Modular Plasmids (JUMP): A flexible and comprehensive platform for synthetic biology

L0 - L2 workflow:

L0 Parts are single parts in a plasmid (promoter, rbs, cds, terminator)
L1 Parts are full assembled transcriptional units(TUs) in an L1 Plasmid (reporter gene and usually kanamycin resistance)

L2 plasmids are four L1 parts assembled using a second type IIS restriction enzyme in an L2 plasmid, using fusion sites ordered in L1 plasmids as A through D.

JUMP Plasmids also contain BioBrick Prefixes and Suffixes if only two to three transcriptional units are required → We require 2 TU's. Downside? Kits are not produced anymore

Due to the presence of a BioBrick prefix and suffix on either side of the main module, two transcriptional units can be ligated together in an ordered fashion using 3A assembly. 3A assembly requires three vectors. One containing the first transcriptional unit with a unique antibiotic resistance gene. Another plasmid with the second transcriptional unit and a unique antibiotic resistance gene. And finally the acceptor plasmid with its own unique antibiotic resistance gene.

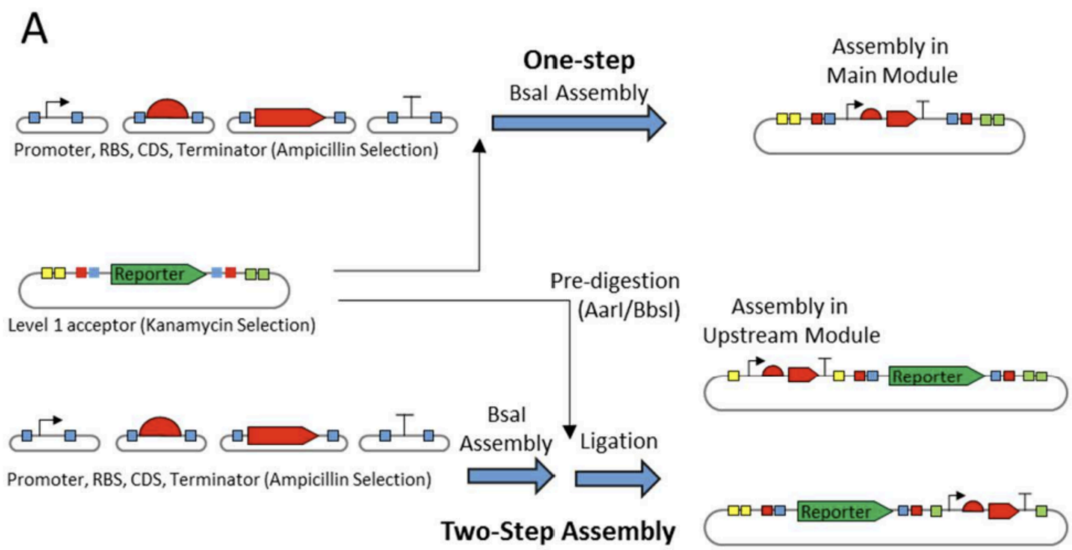
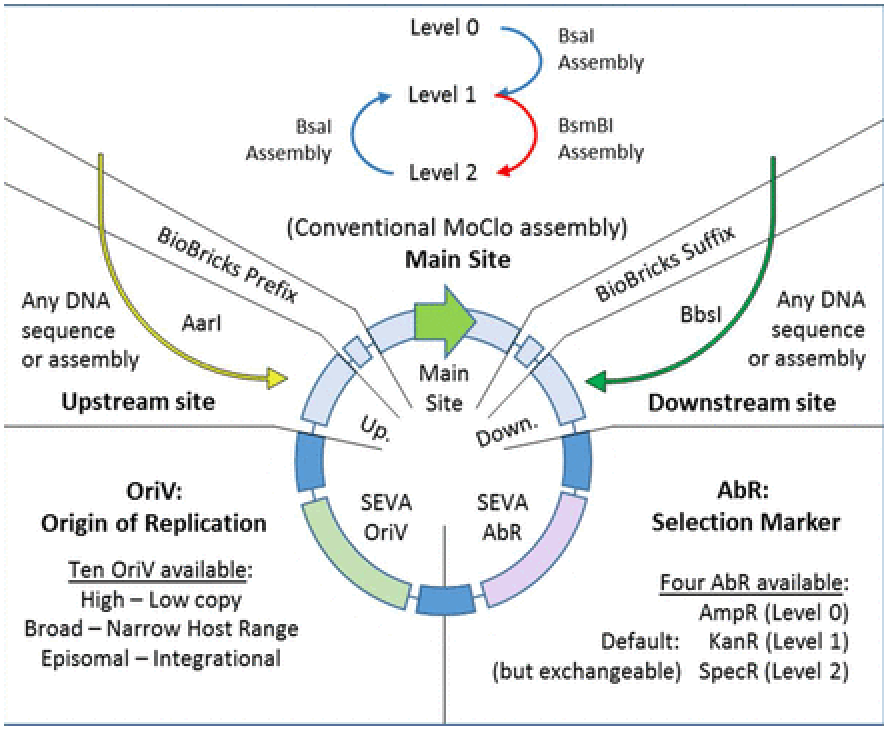


Figure 1. Modular cloning and PhytoBrick standardisation of parts. 1A) Basic parts are contained in level 0 vectors, which are assembled to form a single Transcription Unit in a level 1 vector. To simplify cloning and screening, the destination vector contains a different selection marker than insert-donor plasmids and a reporter gene replaced by the assembled TU. A level 2 vector can combine multiple level 1 assembly products, due to the use of an alternative type IIS restriction enzyme and selection marker between different assembly levels. In JUMP level 1 plasmids can be used as level 3 assembly destination vectors. 1B) PhytoBricks standard for fusion sites (18); format of JUMP parts provided in toolkit; fusion sites of level 1 and level 2 JUMP vectors.

What are the secondary upstream and downstream sites? :



Secondary sites are separate type IIS sites. Not necessarily relevant as enzymes AarI and BbsI don't come in kits from NEB.

Main question of meeting: If we need matching fusion sites in an L2 plasmid (for multiple transcription units) but not as many as 4, how can we save time?

Questions for Matt:

1. Why can't we print out each TU and insert into the L2 with one restriction-ligations?
    a. What do we lose by not using the modular parts in the distribution kit? Is the time saved not worth a potential risk with expression that we are not able to see?

7/8/2026 Bile Acid Promoter Lit Review: Humza (5 Hrs)

Article: https://link.springer.com/article/10.1007/s00253-019-09743-w#Tab4 **Figures + Content:**

"The influence of several concentrations of the tested bile compounds was analyzed in all the strains transformed with pNZ:16090-aFP. In concordance with the results with bile salts, none of these compounds produced induction of fluorescence in *L. lactis* and *Bifidobacterium* strains, while the *Lactobacillus* strains showed different levels of induction with each compound. In general, the fluorescence increased with the concentration of the bile compounds, although not in a linear manner and restrained by the growth-inhibiting effect of the compounds.

Lb. casei BL23 pNZ:16090-aFP showed stronger induction with cholic acid than with any of the other compounds, when the lowest concentration (0.025%) was used (Fig. 3a). The two *Lb. rhamnosus* transformed strains displayed similar results to *Lb. casei* BL23. Cholic acid concentrations higher than 0.025% affected negatively the growth of these three *Lactobacillus* strains. Other compounds did not cause such strong growth inhibition, and allowed the comparison of induced fluorescence among the different bile compounds, using in each case the highest concentration that allowed the growth."

Home > Applied Microbiology and Biotechnology > Article > Table 4

Table 4 Fluorescence mean values obtained from *Lb. casei* BL23 transformed with plasmids containing each promoter and grown with or without bile salts

From: Bile-induced promoters for gene expression in *Lactobacillus* strains

	Bile salts concentration (%)		
Promoter	0	0.1	0.2
P16090	0.006 ± 0.006^{aA}	0.565 ± 0.018^{bB}	1.118 ± 0.041^{cA}
P0779	0.151 ± 0.012^{aB}	0.481 ± 0.030^{bA}	0.779 ± 0.023^{cB}
P2056	0.434 ± 0.054^{aC}	0.612 ± 0.016^{bB}	0.826 ± 0.040^{cB}

[a–c] Different superscript small letters within the same row indicate significant ($P < 0.01$) differences between fluorescence obtained for a given promoter under different bile salt concentration by Tukey test

[A–C] Different superscript capital letters within the same column indicate significant ($P < 0.01$) differences between fluorescence of the different plasmid tested with a given bile salts concentration by Tukey test

Notes:

- P16090 responds well to cholic acid induction. LZU-CHINA iGEM team shows successful usage in BL21 strains for Deoxycholic acid biosensor. It is important to note that 0.05% concentration of cholic acids was the safest concentration of Lactobacillus growth, as LCA is STILL cytotoxic at certain concentrations

DBTL 2 Literature Review: L0 -> L2 Assembly and Methods + Promoter Experimentation

- Human fecal samples were taken in order to test for gut-based activation
  - "In order to test bile induction under simulated intestinal conditions, Lb. casei BL23 pNZ:16090-aFP was co-cultured with human fecal microbiota with and without bile salts. Interestingly, the expected fluorescence induced by bile during the co-culture was maintained over time, during the growth on MRS plates, although they did not contained bile."
- It seems that the promoter is Lactobacillus specific and shows no activity characterization for Bl21.

Article: Bile Acid Biosensor (ACS)

Figures and Quotes:

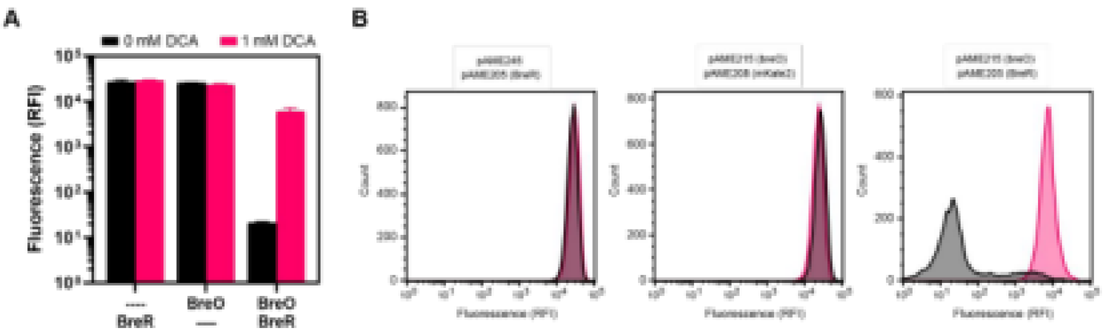
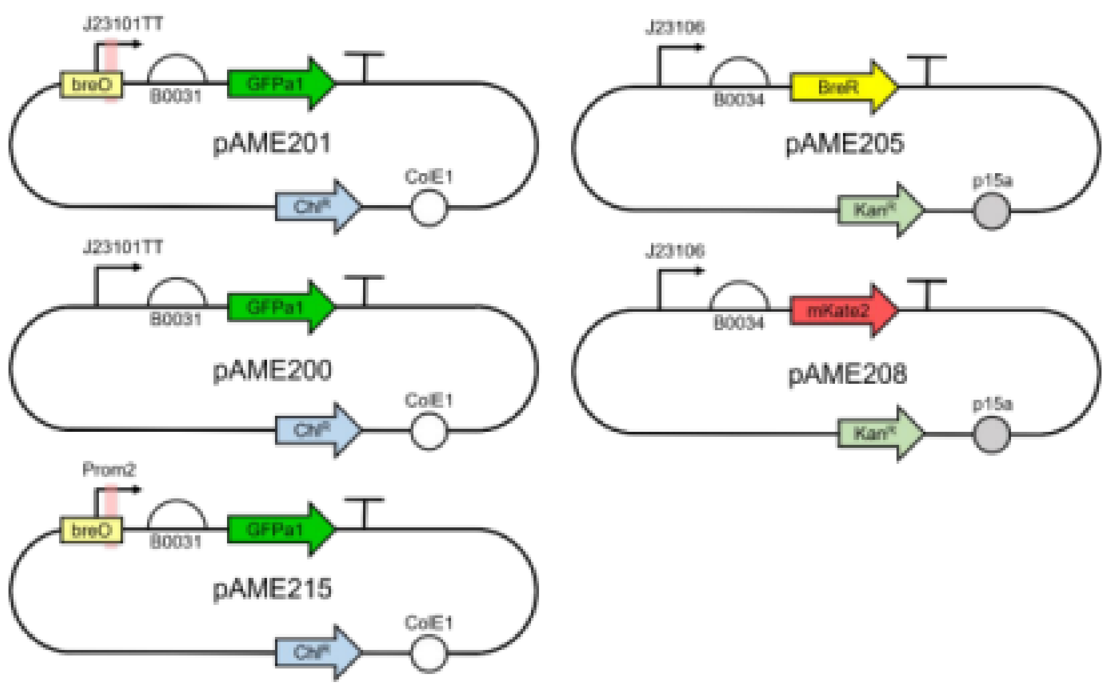


Figure S6. Sensor Controls. A – Fluorescence levels produced by *E. coli* strains carrying a plasmid expressing GFP under control of promoter Prom2 with or without the presence of the breO operator (pAME245 and pAME215, respectively) along with a second plasmid expressing either BreR or mKate2 (pAME205 and pAME208, respectively) in the presence and absence of 1 mM DCA. Experiment was run in triplicate with mean and standard deviation shown. B – Representative flow cytometry histograms corresponding to the data in panel A.

DBTL 2 Literature Review: L0 -> L2 Assembly and Methods + Promoter Experimentation



BreR (full “34_BreR_rnpB” synthetic block, BreR in bold)	TCTAGAGAAAGAGGAGAAATACTAGACTAGT**ATGAAACTG** **AGTGAACAGAAACGTCTGGCCCTGATTGAAGCAGCCAAAG** **AAGAGTTCACCCAGTTTGGCTTCCATGCAGCCAACATGGA** **TCGTGTTTGTGAACGTGCCGGTACAAGCAAACGTACACTG** **TATCGTCACTTTACCTCCAAAGAACTTCTGTTCATTGAAG** **TGATCAATCTGCTGGTTGCACAGCCTCACAAAGTTGGTTT** **CGAGTATCAGTCAACCCGTAGTCTGGCAGATCAGCTGCAT** **GATTACTTTGCAGCAAAGATTGATCTGCTGTATCGTACCA** **TTGGTCTGGATGTTCTGCGTATGATTGTTGGTGAGTTTGT** **TCGTGATCCGGCACTGACCCAGCAGTATCTGGCACTGATG** **GGTACACAGGATACCGCACTGACCGCATGGCTGCAGGCAG** **CCATCAAAGATGGTAAACTGATTGAGAAAGAAGTTGCACC** **GATGGCAACCACCCTGATGAATCTGTTTCATGGTCAGTTT** **CTGTGGCCGCAGCTGCTGGCAGCAGTTGAACTGCCGGATG** **CCAAACAACAGCAAATCATGATGGATGAAATCATTCGTGT** **GTTTGTTCTGAGCTATGGTGTTTCAAGTCCGAGTCATCTG** **AGCATTGAACTGAAACCGTAA**GAGCTCCCGGCTTATCGGT CAGTTTCACCTGATTTACGTAAAAACCCGCTTCGGCGGGT TTTTGCTTTTGGAGGGGCAGAAAGATGAATGACTGTCCAC GACGCTATACCCAAAAGAAAGCGGCCGCTGCAGTCCGGCA AAAAAGGGCAAGG	This study

DBTL 2 Literature Review: L0 -> L2 Assembly and Methods + Promoter Experimentation

Notes: Looking into the plasmid preparation, the BreR construct is a CDS for a transcription factor that represses the BreO operator, coming before the corresponding Anderson promoter in constructs like pAME215. The BreO operator sequence was NOT provided. I will be looking into that further.

Update: I found the following for the construction of the pAME201 plasmid(see above) :

“The original PCR product was separately reamplified with primers AME1000/AME1002 and cloned along with a PCR product containing the breO sequence and promoter (product of a no-template PCR using primers AME1005/AME1004) into J64100 at EcoRI/NotI using Gibson Assembly to create plasmid pAME201.”

Promoter sequence for pAME201(should be complementary to

primer): TTTACATTTAGCTCAGTCCTAGGTATTATGCTAGC

Complement: 5‘-AAATGTAAATCGAGTCAGGATCCATAATACGATCG-3’

AME1004:

3’-AAATGTAAATCGAGTCAGGATCCATAATACGATCGAGATCTCAGTGTGTCCTT TGGATGA-5’ AME1005:

5’-TAGCTTTCGCTAAGGATGATTTCTGGAATTCGGATCCAATGTAAACCCGAGAGTTTACATTTAGCTCAGTCCTA-3’ AME1005 Contains an EcoRI site. Between that and the promoter complement is the breO operator: GGATCCAATGTAAACCCGAGAG

Possible Experimental Routes + Plasmid Assemblies:

1. Separate insertion of BreR into JUMP Plasmid, use of breO operator to limit production of BOTH Cassettes
2. Anderson promoter usage with equal strength across BOTH Cassettes
3. Use of BreR + breO for just KIATPSL + PcAPSK, anderson promoter for mBT-SULT
4. Positive Controls(these would be two different constructs): Anderson promoter/BreR + breO under sfGFP
5. Negative Controls: breO promoter used WITHOUT BreR transcriptional unit with sfGFP (See Figure S6)

Note: Not all can be used, we have to pick and choose, since we also have to vary concentrations of LCA for induction/sulfonation.

Action Items:

7/15

Coordinate the necessary golden gate assemblies for L0 -> L1 with references to the identified distribution kit for our operator characterization. Essentially name the steps required for the assembly of the four plasmids using the JUMP vectors using the necessary parts from the distribution kit. We can likely integrate parts like Ribosome binding sites with promoter sequences so combine those for now. Note the matching 5' and 3' sites though so the golden gate goes smoothly. Goal: Use sfGFP expression to measure expression strength using standard anderson promoter (choose an anderson promoter, we can get more printed. I would reccomend the one with strength 0.51) Plasmid assemblies: Promoter + sfGFP, Operator + Promoter + sfGFP, Promoter + sfGFP AND second plasmid with Transcription Factor, Operator + Promoter + sfGFP AND second plasmid with Transcription factor

Look into the necessary experimental constructs for the actual gene expression? If we first characterize and figure out the expression of sfGFP under the influence of the operator and transcription factor then we can weed out some of the construct combos for the full circuit w/two cassettes. What I need ygs to figure out is what is most optimal for an experiment. The first two things on my mind rn are plasmids where we have the operators present in both cassettes and one where we don't have the operator present at all and we have the Anderson promoter on its own. We can also vary our usage of the transcription factor in a separate plasmid for experimental purposes. Go ahead and look into designing that experiment pls.

Terms:

- sfGFP: coding sequence for superfolder Green Fluorescent Protein (trated as level 0 gene/CDS) - serves as a reporter of expression strength

Four plasmids:

1. Promoter + sfGFP
2. Operator + Promoter + sfGFP
3. Promoter + sfGFP and second plasmid with TF
4. Operator + Promoter + sfGFP and second plasmid with TF

Steps:

1. level 0 parts for the TU's: operator part, anderson promoter (from kit, strength 0.51), RBS, sfGFP CDS, Terminator
2. Find these parts in the kit, making sure they are compatible with BsaI based golden gate for L0 to L1
3. For sfGFP reporter plasmid: mix backbone and all of its L0 parts in a BsaI + ligase reaction, so BsaI cuts out recognition sites and have ligase join parts
4. For TF cassette plasmid: combine backbone + promoter + RBS + TF CDS + terminator with BsaI + ligase
5. You get:

A alone (Promoter + sfGFP)
B alone (Operator + Promoter + sfGFP)
A + C (Promoter + sfGFP AND TF plasmid)
B + C (Operator + Promoter + sfGFP AND TF plasmid)